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Sustainability Transition in Forestry Transport through Electrification: Perspectives on Business Model Development

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ABSTRACT

The electrification of heavy-duty transport is a critical pathway towards reducing emissions in Sweden's forestry sector, a major contributor to national CO2 output. However, the adoption of battery electric trucks in this context remains limited, and the transition poses significant challenges for existing actors and newcomers. This paper investigates how diverse actors within the forestry transport ecosystem interpret the electrification transition and its implications for business model development. Drawing on empirical data - comprising seven experimental sites, two workshops, and 17 stakeholder interviews - this study foregrounds the role of interpretation and sensemaking in the early stages of a complex sustainability transition. The findings reveal that electrification is perceived not as an incremental change but as a radical innovation requiring cross-actor learning, new partnerships, and ecosystem-level coordination. Actors face strategic and operational uncertainty, particularly around value capture, data sharing, and infrastructure ownership, while also engaging in collaborative exploration of new roles and opportunities. By highlighting the interpretive processes shaping business model adaptation, this study contributes to the literature on sustainability transitions and business model innovation. It also offers practical guidance for managers aiming to navigate uncertainty, foster collaborative innovation, and co-create value in evolving industrial ecosystems.

KEYWORDS

Sensemaking, sustainability, innovation

INTRODUCTION

The energy use in the Swedish forest sectors' domestic road transport is estimated to account for 25% of all commercial road transport (Fossilfritt Sverige, 2020), and domestic road transport accounts for roughly a third of Sweden's national CO2 emissions (Naturvårdsverket, 2024). Consequently, the transition toward sustainability and net zero demands unprecedented shifts. One promising pathway, electrification, is currently gaining interest and traction among industrial firms and policymakers. Battery Electric Trucks (BETs) are widely recognized as key enablers of climate-neutral heavy-duty road transport (Shirizadeh et al., 2024). Yet, their adoption in the forestry industry is very rare to date. A transition to electrified forestry transport requires not only technological innovation, but also the transformation of business models, operational structures, and new collaboration to be viable.

When it comes to business opportunities specifically, electrification introduces both challenges and opportunities for business model innovation, impacting existing and emerging actors. New areas of competition include smart charging, energy storage, and frequency balancing, which adds complexity in value capture and increases uncertainty for businesses (Audy et al., 2007; Darwish et al., 2022). Meanwhile, existing actors in the ecosystem have a powerful position to influence how the transition takes shape (Geels et al., 2017; Lind & and Melander, 2023). The transition consists of many different system levels, including new technology (e.g., battery-powered trucks, charging solutions, data solutions), acceptance of stakeholders (e.g., customers and drivers), and reliance on policy and infrastructure (e.g., political decisions on tariffs and reduction obligations, charging infrastructure). Altogether, this suggests that several actors need to change to facilitate systems change (Geels, 2012).

In the ongoing demonstration project "TREE" (TRansition to Efficient Electrified forestry transport), a wide range of industrial actors demonstrate the use of electrified trucks in the forestry industry, contributing significantly to industry-wide reduction in CO2 emissions. The TREE project presents a unique opportunity to study seven geographically distinct experimental sites (i.e. networks), each with varying operational setups and actor networks. These networks evolve differently over time as actors engage in orchestration and self-organization. Understanding the dynamic between these actors is critical to fostering the successful development of business models in this transitioning industry.

At hand, the electrification of forestry transport is a highly complex and developing phenomenon, with many uncertainties and risks that directly affect business model viability which need to be better understood. Consequently, the present paper aims to shed light on how different actors involved in the electrification of the forestry industry interpret the electrification transition and its implications for developing each actor's business model. As multiple actors with distinct goals and logics are involved in interpreting and responding to this transition, understanding how they make sense of the change becomes crucial for analyzing how business models are developed and adapted by each actor (Haffar & Searcy, 2019; Weick et al., 2005). This is crucial to avoid potential barriers to industry transformation, including conflicting interests, imbalances in value capture, and inadequate risk-mediating mechanisms.

The data analyzed in this paper comes from two main sources: (1) documentation from two workshops involving over 50 participants from organizations involved in the transition of the industry and (2) 17 in-depth interviews with key stakeholders across the project sites in the early stages of their electrification transition. Analysis of this data will provide rich qualitative insights into how these actors interpret their role in the transition as they work to align their business models with the electrification goals of the forestry industry.

The analysis will provide new insights into business model innovation at the industry level, during a sustainability transition. The findings will contribute to the academic literature by offering a detailed examination of how value is created, captured, and shared across networks in the context of electrifying transport. Furthermore, this research will offer actionable guidance for practitioners, enabling them to understand and manage an industrial sustainability transition more effectively.

THEORETICAL BACKGROUND

Business model development

Research on business models and their development has been constituted a major stream within business and management literature for over three decades (Zott et al., 2011). Consequently, the business model has been described in numerous different ways but typically can be condensed into three elements: the value proposition, the value creation and delivery, and the value capture (Osterwalder et al., 2010). Here, the value proposition describes what value the company creates and delivers to its customers, i.e., what needs they are satisfying and what customer problems they are solving. It is well established that technological innovation (such as the electrification of heavy transport) can create a need for business model development. Teece (2010) highlights that the more radical the innovation and the more complicated the revenue structure, traditional business models will likely require changes. Furthermore, the new business model requires a process of discovery, learning and adaptation as they are not likely to be self-evident or plug-and-play in established organizations. This may be especially true for established organizations that are slower to change following established processes, values, and culture (Teece, 2016). For this reason, Teece (2010) suggests that experimental business models must be tested with the current state of the business ecosystem, and how this ecosystem may evolve over time must be considered.

Furthermore, Hedvall et al. (2019) studied how companies in the forestry industry (i.e. truck OEM, transportation providers, and transportation buyers) move from working as standalone firms with single offerings in supplier-customer dyads, towards offering multiple *solutions*, taking on multiple roles in business networks, and where different solutions are interconnected in networks of solutions. This perspective on offering solutions through business networks is based on the industrial network approach which puts emphasis on processes for the co-creation of value and servitization (Hedvall et al. 2019). While business development within business networks is inherently complex due to the involvement of multiple interdependent actors, this complexity is further amplified in sustainability transitions. In such contexts, actors must also manage new types of uncertainties and risks. These elements often extend beyond traditional business concerns and increase collaboration and learning together to ensure environmental, social, and economic objectives.

Sustainability transitions and business model development

The research on business model innovation for sustainability has surged in recent decades, as the business model concept requires a systemic perspective and is seen as helpful in connecting insights at the different levels of analysis relevant to sustainable innovation (Boons & Lüdeke-Freund, 2013; Stubbs & Cocklin, 2008). Business model innovations for sustainability have been defined by Bocken et al. (2014, p. 44) as: “innovations that create significant positive and/or significantly reduced negative impacts for the environment and/or society, through changes in the way the organization and its value-network create, deliver value and capture value (i.e. create economic value) or change their value propositions”. Here, value is seen from a sustainability perspective (Evans et al., 2017), including environmental (e.g. renewable resources, low emissions, low waste, pollution

prevention, etc.), social (equality and diversity, well-being, community development, etc.), and economic value forms (e.g. profit, return on investment, business stability, long-term viability).

Geels (2017, p. 474) further emphasizes that in many system transitions relating to sustainability, actors struggle to rely on rational or technical decision-making following the inherent complexity of these transformations. Therefore, the decision-making process must account for multiple uncertainties, including "future costs and performance of radical innovations, social acceptance, consumer interest, and policy support". Additionally, "disagreements regarding desirable solutions, policies, costs, and benefits, as well as the distribution of power" among stakeholders further complicate these transitions. As such, relying exclusively on technically rational criteria for decision-making is inadequate. Instead, discussing and navigating the process collaboratively is necessary, ensuring that all stakeholders contribute to and align with the transition goals. Therefore, navigating risks and uncertainties related to developing new business models during technology transitions is highly interesting.

Business model development when transitioning to electric trucks

Technological innovation and decarbonization are having a clear impact on road transportation systems (Geels, 2012). In this change, traditional business models of incumbent firms may be threatened, and new entrants with complementary innovations (e.g. batteries, smart grid solutions, sensors, software solutions for accessing charging infrastructure, etc.) bring new business models to the table (Geels et al., 2017). Electric adaptation of trucks comes with the challenge of simultaneously optimizing and developing the charging infrastructure (Alp et al., 2022). The development and the decisions for how to invest in charging, what actor should invest in it, and its scaling are coupled. The charging infrastructure can be built and operated by both new actors who specialize in that or by actors in the established transport system, such as forestry companies or haulers.

Truck manufacturers that traditionally rely on truck sales and preventative maintenance now require new collaboration partners, new resources, and relationship-building with actors in their business network for future viability (Lind & and Melander, 2023). Furthermore, they outline managerial decisions relating to using their power to either create 'lock-ins' that favor the current system, leaving the present system and investing heavily in the new system to take a leadership role, or trying a combination of the two alternatives at risk of being caught in the middle, pointing to a need to work ambidextrously (Lind & and Melander, 2023; O'Reilly & Tushman, 2013). These new business networks can include constellations that previously considered each other as competitors to create value, e.g., by working with competitors to share charging infrastructure (Melander & Wallström, 2023).

Sensemaking of business model development for sustainability

When multiple actors representing different organizations with distinct goals and logics are involved in interpreting and responding to a major transition, understanding how they make sense of the change becomes crucial for analyzing how business models are developed and adapted by each actor. Sensemaking can be understood as a process of organizing where actors rationalize ongoing circumstances by extracting cues and making plausible sense to give the circumstances order. In other words, it is a crucial process for people and organizations to make sense of events, changes in the business environment, and other circumstances to take action based on a certain understanding(Weick et al., 2005). The sensemaking process becomes complex in business and strategy as diverse actors

have different prerequisites that shape how they interpret (or '*frame'*) certain situations. Kaplan (2008) describes how this can give rise to framing contests, where different actors aim to establish the legitimacy of their particular framing to give rise to a dominant framing, leveraging power dynamics. Without the emergence of a dominant framing, divergent framings can lead to conflict and slow down decision-making.

Different framings of sustainability in business organizations have been a salient topic of investigation. Hahn et al. (2010) describe that much of the early management literature on integrating sustainability follows the view of a win-win and business case of sustainability, i.e., that social, economic, and environmental sustainability are mutually reinforcing. They argue that this perspective is problematic for individual actors, as many of the ambiguities relating to innovation for sustainability can be derived from the often-divergent goals of economic, social, and environmental sustainability, as well as complexity in a wider range of stakeholders (Hall & Vredenburg, 2003). For this reason, trade-offs and conflicts between different goals and actors need to be considered to a larger extent. Later research has also argued for a 'paradoxical' framing, which does not target alignment between different sustainability goals but instead allows for conflicting ones, which in turn opens up for larger complexity (Hahn et al., 2014) and issues that can be 'worked through' in an integrated fashion (Luscher & Lewis, 2008). Similarly, Haffar and Searcy (2019) also found that companies that favor a market-led and business case logic may have a too narrow scope of stakeholders in their sustainability decision-making, leading them to an "if-then" reactive, myopic, and reductionist approach. This is in comparison to what they call a holistic logic, i.e. a need to create shared value and system-wide benefits for multiple stakeholders, where sustainability tensions are managed in a more comprehensive fashion. Thus, when multiple organizations work together towards a common goal, it is highly relevant to understand how their diverging views are used constructively for collaborative sensemaking (Seidl & Werle, 2018).

Summary

This study explores how interpretation in the sensemaking processes influences business model innovation during the electrification of forestry transport. Specifically, it investigates how different actors interpret the transition to electric transport systems through their framings and subsequent actions. Based on the theoretical background, the following research question guide the study:

> *RQ: How do different actors in the forestry transport ecosystem interpret the electrification transition and its implications for developing each actor's business model?*

This study contributes to the literature on business model innovation and sustainability transitions by foregrounding the role of *interpretation* in the early sensemaking process of electrification in forestry transport. Furthermore, it provides actionable guidance for managing complexity and fostering collaborative innovation in uncertain and evolving ecosystems.

METHOD

The empirical setting for this research is a national Swedish research project that focuses on the electrification of the forest industry (TREE: Transition to Efficient Electrified Forestry transport, financed by Vinnova FFI [grant 2023-03185]). The project consists of over 20 partners representing a wide range of stakeholders across the entire value chain. These include forest owners and organizations (transport purchasers), vehicle producers, haulers, transport coordinators, charging

infrastructure companies, and route optimization software developers. A qualitative research approach was applied, motivated by the wide range of stakeholders and the novel phenomenon studied. Further, the formulated research question demanded possibilities for in-depth and qualitatively rich descriptions in order for the researchers to empirically capture how different actors interpreted the electrification transition and its implications for their business model. Hence, interviews and observations of workshops were selected as the primary source of data.

Data collection

The main empirical data was collected in two different ways 1) observations and documentation of a workshop within the project focusing on mapping of the different actors involved in the electrification transition, and 2) interviews with 17 informants representing different actors in the project. In addition, the interview guide used for the interviews was validated with the help of a reference group which consisted of individuals possessing good insight into the various types of actors within the project and for the selection of informants, see Table 1 for an overview. It is worth noting that the organizations' role has been labelled for the sake of simplicity, e.g., "Truck manufacturer" does not capture the full extent of their business or role in the network, and there are nuances between the different forest companies and transport companies that are not captured.

50 participants were involved in the workshop from seven geographically based demonstration sites within the TREE-project. During the session, they worked on mapping all actors, their interrelations, and the flows of materials, data, transactions, and information among them, largely inspired by value network analysis (Allee, 2008; Galle & Matti, 2022). The rationale for this exercise was to capture the understanding of the relations between the actors involved in each site. It was also used as material to spur questions, which they discussed together.

After the workshop, the interviews were conducted with selected participants to gain a deeper qualitative understanding of the challenges they faced during the electrification transition, with a specific emphasis on the development of their business models. During the interviews, informants were shown a digital version of the network map related to the sites their organization was part of, developed during the workshop. They were asked to reflect, reason, and discuss business model-related challenges and opportunities. All interviews were held via video calls, lasted between 60 and 90 minutes, and were recorded and transcribed.

The second workshop focused in-depth on the identified challenges and opportunities together with representatives from one of the project sites. In this particular site, several roles (e.g. transport purchasers, transport companies, etc.) were subsidiaries or close partners of the forest company, meaning that more sensitive and strategic business-model questions could be discussed openly.

Table 1: Overview of informants, their role, and the role of their organization in the network.

Informant	Informant role	Organization
A	Business manager logistics	Transport company
B	Logistics manager	Forest company
C	Purchaser, logistics and transport	Forest company
D	Purchasing and transportation manager	Forest company
E	Business developer industrial supply	Forest company
F	CEO	Transport company
G	Sales manager e-mobility	Truck manufacturer

H	Managing partner	Software solutions provider
I	CEO	Transport company; Charge Point Operator
J	Logistics manager	Transport company
K	Researcher forest operations business	Expert
L	Researcher	Expert
M	Co-founder, CEO; Charging point expertise	Expert
N	Project manager	Expert
O	Researcher	Expert
P	Transport service manager	Expert
Q	Business developer	Expert

Since the research question concerned the commercial actors, it was decided that the analysis for the present paper should focus on informants A to J, and the remaining interviews served as experts and provided a greater understanding of the industry for the research team.

Data analysis

The interview transcriptions were read through and interview recordings revisited. Initial codes relating to challenges, overall reflections concerning business model development, and electrified transport were created descriptively and in vivo (Miles et al., 2014) to capture the participant's views. The initial set of codes was then collected in a spreadsheet together with the full quotes. Higher-order "themes" were created by looking for patterns in the data by sorting and counting recurring codes. To give further structure to the analysis, the business model elements of value proposition, creation, delivery and capture were utilized. The participant's reflections and views were then re-formulated as questions to describe present concerns and key questions in business model development.

RESULTS

The participating informants were grouped into five categories based on their primary role in the forestry transport value chain: forest companies (transport purchasers), transportation companies (i.e. transport coordinators and haulers), truck manufacturers, charging point operators, and software solutions. An overview of the identified need for business model development and key questions concerning value proposition, creation, delivery, and capture, for the electrification transition for each actor is shown in Table 2 below.

Table 2: Identified need for business model development and key questions concerning value proposition, creation, delivery, and capture.

Actor	Need for business model development	Value proposition	Value creation	Value delivery	Value capture
Transportation companies [A, F, I; J]	Current transport offerings include the fuel price - possible that future offerings should exclude this.	When will customers no longer accept diesel transport and request green transport? Can we offer transportation without fuel/electricity? Can we offer good transport solutions - not just transport? Can we use the BETs for frequency balancing?	How to engage competitors in the transition? How to optimize complex flows with shorter range and need for charging? What role does digitalization have in operations?	How do we build and maintain long-term relationships with customers over transport solutions? How can we increase the utilization rate of the BET?	When is it no longer viable to invest in diesel trucks? How to price transports for multiple customers? What is a suitable depreciation period of BETs? How to manage the large investment in BETs? What are the additional costs associated with the BET?
	Current business model has too little incentive for creating good transport solutions - shorter range creates need for better solutions.	When will customers no longer accept diesel transport and request green transport? Will future competition be on lead time, sustainability, or both?	Can our customers offer charging solutions and cheap electricity? When will other transport systems transition to electric and how does that impact us?	How can we increase the utilization rate of the BET?	When will the price of BETs decline? Who will cover the increased cost of transportation? Will consumers accept higher costs of end-products? How will energy prices fluctuate?
	Current business model has too little incentive for creating good transport solutions - shorter range creates need for better solutions.	When will customers no longer accept diesel transport and request green transport? Can we offer good transport solutions - not just transport?	How can we be less reliant on the expensive public fast-chargers?	How can we increase the utilization rate of the BET?	How will energy prices fluctuate? What are the running costs of operating the BET?
Charging point operator [I]	Relatively newly founded company	How can we offer full charging solutions?	What new competence is required in offering charging solutions? How do we engage the energy suppliers, local authorities and policymakers?		

Actor	Need for business model development	Value proposition	Value creation	Value delivery	Value capture
Forest companies [B, C, D, E]	Other actors are more affected than transport purchasers, particularly route planners.	How can we offer charging solutions? How can we take some of the risk to support transport companies?	How can we achieve transparency with partners? What are our own goals in relation to electrified transport? How can the large actors influence charging operators?		How can we take control of a larger portion of transport costs? Can we offer charging?
	Business model during and after the project will likely differ. Challenging if business models in the industry differentiate too much.	How can we help our customer offer fossil-free products? How can we offer charging solutions?			How to deal with the difference in charging costs between public and private charging solutions? Can we offer charging and how would we deal with transporters carrying competitor loads as well as ours?
		How can we offer charging solutions?	How do we balance openness with oversharing information with competitors?	How do we build and maintain long-term relationships with all actors involved in the electrification?	
	Need to rethink the whole process and system. Not just exchange one transport system with another. To make it effective will affect the industrial process. What is our role in the charging infrastructure?	What do buyers of transportation value? Should electrified transport to be compared to diesel or are there new values in sustainability? Can electrification make transportation cheaper?	What partners do we need in electrification and how do we cater to the right customer? What knowledge do we need to make the electrified transport efficient? How should our contracts look like?	What is the role of alternative fuels?	How do we calculate compensation and formulate contracts? Who is willing to pay for the risk?
	No long-term interest in owning trucks, offering charging, or frequency balancing. Will experiment with setting up their own charger out				When will the price of BETs decline? How will haulers be able to cover the investment costs of BETs? Will consumers accept higher costs of end-products? How balance incentivizing

Actor	Need for business model development	Value proposition	Value creation	Value delivery	Value capture
	of necessity, unsure of if/how partners should pay for this.				efficient transport and sharing cost of charging with partners? How to counterbalance fluctuations in electricity prices? Should we charge our transport partners for charging at our sites?
Truck manufacturer [G]	How can new business models minimize risks for all stakeholders?	What is the potential of BETs in forestry transport? Can we offer new flexible charging solutions to accommodate for the urgent needs of forestry transport? How can we offer charging solutions? How can we help customers finance the transition? What other applications do the batteries have?	How do we influence policymakers in supporting the electrification? How do we balance openness with oversharing information with competitors?	How can we get transport purchasers to require green transport?	What is the residual value of the BETs after leasing? What is the value of the used batteries? What is a suitable leasing fee for BETs? How can we decrease the cost of production?
Software solutions provider [H]	Sees that the transport industry is shifting. Risk that powerful actors create lock-ins and leave out third-party actors.	What digital solutions aid in electrified forestry transport? How can we use data in a smart way?	How can we get the data required for our offerings? How can open data standards and protocols be used?	How can we avoid potential lock-ins in the industry and promote cooperation?	Which of our tailored solutions can be scaled?

Looking at Table 2, most informants reflect upon value capture, i.e. the economic feasibility of the electrification. In comparison, few discuss aspects related to value delivery, which is to be expected given the early stage of the transition. In the following section, the informants' interpretation of the how electrification of forestry transport changes the value proposition of different actors, and how this value is created, delivered and captured, is further elaborated on.

Value proposition

Regarding the value proposition, some informants in transportation companies reason about whether fossil-free transportation brings additional value to their customers or the customer-of-the-

customer or not. While the current customer demand for fossil-free transport is seemingly low, several of the informants see a point in time where the demand for fossil-free transport will outgrow the acceptance of diesel-based transportation, meaning that the investment in diesel-trucks should also be carefully considered in terms of long-term viability.

> Do you really dare to invest in a diesel truck […] when the transition will go so fast? When do you start risking more investing in a diesel truck than you do when you invest in an electric? For a customer could demand that "we only want electric" or "minimum 70% electric" and suddenly, you have loads of diesel trucks still on their depreciation period. – J

Interestingly, one informant describes that many forestry transportation companies closely follow the development in e-commerce where transportation companies compete on lead-time, and suggests that a future competitive advantage could be environmental sustainability instead.

> Many people look at e-commerce, they compete with good lead times. […] And of course, in its own way it is madness to keep on being fast instead of perhaps optimizing on being environmentally friendly. But it's not always necessarily opposites but can go pretty well hand in hand - A

Several informants among transportation companies see that the main value proposition of transporting goods will remain after electrification. However, there is an interesting nuance between informants. There are those who see that battery power is simply an alternative fuel comparable to diesel. Contrastingly, others see the constraints of lesser range as an opportunity to rethink if this is an opportunity to move more into offering transport solutions rather than transportation.

> No, I don't think there [is any major difference in business logic] but... If it were a little cheaper or cost-neutral [with electric], we have a lot of transports that we could change, a huge amount. - A

> I believe that we will be more of a knowledge partner for transport buyers than just being a transport company, that we deliver a more of a complete solution […] That means discussing solutions with the customer to make their transport price as cheap as possible. Not just driving. – J

> What I'm hoping for, speaking of business models and pricing. We can hope that the limitations on the vehicle's range may contribute to people thinking a little differently […] when you suddenly get tougher demands, you need to think a little more. - F

Notably, given the non-existence of electric vehicles in Swedish forestry transport prior to the present project, the involved transportation companies currently have a unique position and first-mover advantages, where they can use their knowledge to offer guidance and solutions to early adopters.

> If you look at electrification, [we can] do business in a different way. We can present a value proposition that perhaps not everyone can. […] it creates new business for us in that we are in the market for those who want to try electrification. And since we are as early ahead as we are, we get a lot of questions from customers that about if can we electrify their flows or if can we help them calculate. There have been customers who are not at all [local] but very far from here. – J

There are also new actors in the project with a clear value proposition in line with electrification, specifically offering charging or software solutions that ease the added complexity of electric transports. These are also companies with relatively unique capabilities and in a niche market to date.

> We design the whole kit and that is a pretty unique skill. In addition, we do the whole project, acquire a piece of land, develop energy together with the electricity network companies […] If you would like to have a charging park, we can design it and hand over the key to you. So the [we] have a rather unique expertise. – I

Considering charging, this is also an area where the forest companies are divided regarding value proposition. Specifically, most of them are installing or considering installing charging solutions in the proximity of their facilities for the BETs. Interestingly, the forest companies generate a significant amount of energy mostly using wind- and waterpower. However, there are differing views on whether the charging stations are a short-term solution out of necessity to make their own electric transports economically feasible, or if it could bring long-term value.

> If we could offer charging infrastructure on our sites ourselves, we would sort of own that [cost] more than we do today [...] Then, there should be a business model around that as well, sustainable at all levels […] It doesn't take a lot of wind turbines to meet all our transport needs. - B

> Charging stations are like, well, we haven't really decided on how we think about ten years from now. But to build your own charging stations or not, basically, we have never built any gas stations either. […] It's not something that we see that we want to do either. But maybe, we must [provide our own] for it to work. - E

Regardless of whether providing charging is a temporary solution or a long-term business, all informants agree that this new value proposition is associated with many challenges and dependencies on policymakers, power companies, and demands new business models.

> But it's also very uncertain in the long run, different tariffs and all this stuff, power consumption and things like that, with charging infrastructure. […] There is nothing stated that it should be our business to sell electricity on our sites, it is rather my own reasoning if you could make money on it, we could keep down the transport cost, which is still a pretty significant part for us. - B

> If we have a haulage company that we have an agreement with […] we provide power from our own production [...] well, what should we do when that truck carries load for [our competitor] then, for example? How would that look like in our business relationship? Can we even do that? - C

When it comes to the value proposition for the truck manufacturer, the potential of BET in forestry transport applications is the main point of interest. However, within the project, they are also demonstrating the use of a new electrified trailer, as well as a portable charging solution. The electrified trailer could provide a significant performance increase to electrified transport and could also be valuable as a hybrid solution with a diesel truck. The portable charging solution could be particularly valuable in transporting logs, as the pick-up locations vary significantly, and the charging infrastructure is scarce. In addition, given the large investment cost for the BETs, the truck manufacturer also provides financing solutions through a subsidiary. Lastly, they are active in providing stationary charging solutions and software solutions for making transportation more efficient as well, making them a potential competitor to several other actors in the project.

> Among other things, it is part of the TREE project as well, we are looking at how to find charging solutions for these trucks […] looking at charging solutions, but also, a form of electrified trailer [...]. And help our customers to electrify their depots, i.e. where you stop with the truck either during the day or overnight […] So that it a completely new business area that we are looking at – G

In summary, the transition to fossil-free transportation is reshaping the value proposition across the forestry transport ecosystem. While current demand may be modest, there is a growing awareness of electrification as strategically important and providing new business opportunities. Transporters are beginning to reconsider their roles, shifting from mere carriers of goods to providers of integrated transport solutions, while new and incumbent actors are entering the space with specialized offerings, from charging infrastructure to software. The first movers in this project position themselves, adapt to the new demands, and stand to gain competitive advantages.

However, doing so requires navigating uncertainty, investing in new capabilities, and rethinking long-standing business models.

Value creation and delivery

Notably, most actors reason about value creation rather than value delivery, which could be explained by the early stage they are in. Therefore, these are presented together in this section.

Many informants describe the added complexity of electrified fleets. The complexity is due to a combination of many factors including shorter range, lesser ability for taking return loads, availability of charging solutions, stand-still time required for charging, etc.

> […] you have to add energy calculations, you have to support with charging capacity, you have to charge at the right time, and then also if you go on to frequency trading and power demand tariffs... It's... It will be very complex to steer such a company forward... If you want to be good. And then also think about the fact that we will produce our own energy as well. The electric business will have a completely different content than the diesel business has had. – I

These challenges combined, informants argue for a greater need for digitalisation than previously to create and deliver value, noting that a significant portion of the forestry transport industry still requires a lot of hands-on manual work as well as pen-and-paper.

> A completely different business, and much more complexity in the business and needs a greater content of digitization to then be able to support that business. […] We have an IT department where we have hired AI-skilled people to support much of the electrification work as well, because we see that we are done with handling this in Excel sheets, we must build robots. – I

> It will be a completely different complexity that is not impossible to solve, and I think it can change the way we have done digitalization, trying to find computer systems and so on that support our transport managers, for example, or support the driver. We may be able to plan more automatically because we are forced to digitize so much in this transition that we must take the next step in the planning work as well, so we may be able to reduce administrative costs in the long run. - J

Furthermore, to deal with the identified challenges, many informants also emphasize the need for partnerships to strengthen the need for new partnerships to gain the right capabilities. Informants point to energy companies, customers, and even competitors in order to succeed with the transition.

> We needed to have expertise and electricity grid competence, as well as with high voltage. […]. Then we chose to ally ourselves with [local energy companies and potential competitor], simply because we didn't want to fight in putting up [charging] sites as well, but it was better to have them with us than to have a split in the market**.** Yeah, instead of fighting I suppose. – I

> What would be good is if [the customer] establishes their own charging station, I think, because they have access to very cheap electricity. [...]. They produce electricity for themselves to some extent as well. Then it could be win-win. - A

> But if our competitors don't shift, then our transition won't go as fast either, that is, if we shift and need charging infrastructure. […] But if there are more people who join, if there are more people who use the market, meaning the more people who shift, the better balance you will also get in prices. You might get better prices on trucks because […] you lower the risk factor a little. - J

However, some also share their concerns when it comes to sharing knowledge and data with competitors. Specifically, since managing a lot of the complexity relies on data, there is a balance of

what the actors need, can, and want to share, since there are competitors in the project. Furthermore, considering business models in general, there are also certain aspects that are strategically difficult to discuss and specifics that are forbidden by cartel laws.

> We follow the "Open Charge Alliance", and that is where we wish, which everyone who works openly does, that those who want to have their infrastructure closed - we do not work with them. Because we believe in sharing, and we believe in transparency. – H

> In the project, we have help from colleagues to be able to determine what may be sensitive and critical, as well as for future digital product development that we are working on and things like that, so that we do not make mistakes. - G

> It requires that all actors as well as all participants in the project contribute, otherwise it will not work [...] But it is also very important in this context that you do not share information with each other that you should not know and so. - D

When delivering a new value proposition, those forestry transportation companies that are starting to offer solutions describe that their customers are immature with regards to this form of offering, and rather, that there is a challenge in the overall industry being rather conservative when it comes to business models.

> I think that business models in many ways will be relatively similar for the industry, I think quite slow-moving there. […]. When we try to bring up things like "we can lower your cost of transport if you do like this…" [the customer thinks] it's a bit strange, "why would you want to do that?". We see that we won't make a profit in two years if someone else will come and make a lower bid than us. Thus, it's better that we build a good relationship and deliver a good solution. So that we will be partly a transport company, partly knowledge company, in between there somewhere. – J

> I think the business models.... Maybe we should think a little differently? […] But I would say that the whole industry is quite conservative generally. - F

In short, the informants point to the fact that electrified forestry transport changes how value is created, delivered, and perceived for many stakeholders, while still operating in a relatively conservative industry. The transition introduces increased complexity that requires digitalization, data-driven solutions, and the development of new capabilities to manage and utilize these technologies. Importantly, the transition also brings questions of reframing existing value propositions due to new procedures and actors. For example, the role of transport operators could change from merely providing transport to offering transport solutions. Being a first mover is viewed by several informants as a potential strategic advantage while they also bring up uncertainties. The added value for environmentally sustainable transport is also not clear. Furthermore, partnerships are seen as crucial, yet at the same time, issues of trust, transparency, and data handling are brought up as delicate issues in a competitive setting.

Value capture

Notably, reflections concerning value capture are the most discussed by the informants. Most concern the heavy investment required for the BET, fluctuating energy costs, residual value, and need for new forms of contracts so that the associated risk can be mitigated. In the project, several actors are experimenting with calculations to see what new costs arise with the BET, comparing it directly to diesel alternatives with the same business model.

> So far, with the electric business we have taken the old business model and calculated the same setup with an electric truck and then we look at: "OK? What are our increased costs at the moment?"

> and then we put a premium. […] We have learned so much in a year and a half that we see that some trucks already cost less today than the fossil truck does. […] We'll learn eventually, there will be other types of business models that will drive completely different costs as well. - I

> It remains to be seen how running costs stack up, versus production and so on. But the [BET] purchases are a lot more expensive, we already know that. - F

Through the project, there are various agreements in place so that the smaller haulers do not have to take the full financial risk of investing in the BET. However, many informants question how a sustainable solution can be found, how additional costs should be distributed, and how to deal with the fluctuating electricity prices.

> Firstly, it's a lot of capital to tie up for an individual haulage company in a truck like this, the purchase cost is, I don't know, it's not twice as expensive, but it's pretty close. It is a risk that we are nevertheless prepared to take together with them. […] And then if you look at the agreement as such, we will be involved in covering the costs associated with driving electrically as well. - B

> That's what we hope for, as technology advances, prices go down. They can't be twice as expensive or even maybe a little more [than diesel]. Otherwise, we will have real challenge that I'm not sure how to solve if we are going to drive electric trucks on a large scale. The result may be that we will not drive to electric trucks on a large scale, because the hauliers cannot buy them. – E

> Consumers or customers' willingness to pay... It's easy to say that everything should be fossil-free, but it tends to be associated with costs, at least initially. […] If you do not have a fuel index for electricity as the electricity price has been in recent years, then it is extremely difficult. – F

Given the heavy investment required, the truck manufacturer offers different forms of leasing solutions. Informants from the truck manufacturer describe that there are several challenges when it comes to pricing these trucks relating to the relatively low production volumes to date, rising costs from suppliers, and the uncertainty relating to residual value after the initial leasing period. Interestingly, other informants are wary about the significant reliance on leasing models, since it may create lock-ins and give the truck owners more power than smaller businesses.

> Looking at electric trucks, I think it's probably more common with some form of leasing because it's such a high capital cost otherwise. […] Today an electric truck costs quite a lot more than a diesel. […] It may be that the customer uses the truck for say five years, then the truck comes back to [us], then there will be this residual value that you need to manage and think about, which in turn then affects the leasing fee as well. […] How can we reduce costs and thereby then the risk exposure both for the customer and ourselves, then the residual value comes in, other opportunities with the product after end-of-life, and the batteries. How can we reuse the batteries in the future? - G

> […] there is a lot of leasing, and there is leasing per kilometer and things like this. They remove some of the risks of buying new technology for the hauliers, but they probably also take away some future opportunities [...] this can probably affect the future business models. […] Virtual powerplants will be quite large, grid balancing, helping the energy grid [...] offer support services is essential [...] and it may be that the ones who will actually have the ownership […] not the haulage company or the driver, that has those possibilities - H

Compared with the present diesel-based system, there is normally an index that is updated monthly to account for changes in diesel prices, providing transparency between transport providers and transport purchasers. With electric transport, there is a greater difference in electricity prices throughout the day and a greater difference in cost for charging based on who provides the charging and the type of charger. Consequently, this divergence creates a possibility for several actors to

capitalize on charging, as well as creates an even greater need for transparency between transport providers and purchasers.

> The idea of our diesel indexes today is a zero-sum game, no one should make money from it, but it should kind of reflect the fuel-share of the cost. It has its ups and downs, but it's an average of the month is still pretty good, while I see that the price of electricity is something completely different [...] This is something completely new for everyone and I think nobody wants to take the risk. Then it is probably very important that there are good business models from the start. These are transparent business models so that you feel confident that no one loses or earns an unreasonable amount of money on parameters like these that can be influenced - B

> It is a very mature [system] with the fuel, we have a good system for dealing with changes in diesel prices. [...] This is the challenge when it comes to electrification. Maybe I can charge at home for 1 SEK, but in a public place to pay 9 SEK. - C

Interestingly, when it comes to revenue models and pricing, the transportation companies pointing towards the possibility of offering transport solutions and greater logistics rather than point-to-point transport discuss the limitations of the present way of pricing.

> It's like: "How many kilometers do we drive and how much does it cost?" […] the business models are very rigid […] When you have a business model then that is kind of just set based on that "you run A to B and it costs this much" and it's a straight price list.... […] I don't think it's optimal to promotes the haulier or incentivize the transport center to improve or to create logistics. I think it's a little too easy to sit and drive back and forth with this business model. - F

> I think that's the main change in business models, the goods will have to be moved more or less the same, but it will surely change bit by bit as you when you can build good logistics, you can build better solutions and better pricing models. - J

In short, value capture is a key concern among informants, driven by the high cost of BETs, volatile electricity prices, and uncertain residual value. While leasing and other mechanisms for sharing risks offer some relief, there are also concerns about long-term control and fairness. The existing diesel-based pricing models do not translate well to electrified transport mostly due to the cost of charging, highlighting a need for transparent and more flexible business models. Here, several informants see a need to move beyond distance-based pricing towards models that support more efficient and holistic logistics solutions.

DISCUSSION

This study investigates how diverse actors in the forestry transport ecosystem interpret the transition to electrified transport and its implications for developing each actor's business model. Arguably, and as highlighted by several informants, the forestry transport ecosystem has a long history with a fairly stable and uncontested business model logic. Our results show that introducing electrified trucks can be seen as a radical innovation since the ecosystem sees a need for discovery, learning, and adaptations over time as existing systems and processes based on a diesel paradigm are considerably challenged (Teece, 2010). In this present early stage, concerns about value capture (i.e. economic feasibility) are significant, as well as the value different actors bring to electrification. Interestingly, despite the strong drive, many informants question the short-term economic viability and customer demands for electrified transport. Overall, this suggests a longer time horizon and a more holistic consideration for creating shared value in the larger transportation system rather for than individual firms (Haffar and Searcy, 2019), and where the agreed transition goal is a strong motivator among the participating project partners. However, several business model shifts and

strategic opportunities can be identified that complicate the transition as they are coupled with significant risks and tensions, see Table 3.

Table 3: Summarizing identified business model shifts, strategic opportunities, and associated risks or tensions.

Actor type	Business model shifts	Strategic opportunity	Key risks or tensions
Transport companies	Redefine value proposition (green and data-driven logistics solutions, not simply A-to-B delivery) Develop new pricing models (e.g. energy excluded, shared charging) Optimize asset use (e.g. higher utilization of BETs through digital routing and planning)	Differentiate via fossil-free transport Offer low-emission transport logistics to environmentally conscious customers	Cost burden of BETs Reliance on infrastructure Pressure of low margins
Forest companies	Shift from being transport buyers to system enablers (e.g. offering charging, offer risk mediation, new contracts and procurement)	Leverage scale and resources to drive industry adoption	Coordinating partners Sharing infrastructure with competitors
Technology providers (charging, trucks and software)	Move towards solutions-as-a-service (e.g. leasing, charging network solutions) Develop data-driven services (e.g. transport optimization, smart charging) Build platforms for electrified transport	Capture value from new services and network effects (or lock-ins) Potential first-mover advantages as remedies to complex electrified transport	Balancing openness for promoting scaling with proprietary advantage

In this turbulence, the studied actors must rationalize the ongoing circumstances, extract cues, and take action based on their understanding (Weick et al., 2005). However, much in line with what Geels (2017) suggests, the rationalization in this case is severely constrained by uncertainties (e.g., fluctuation of electricity prices, market demand, residual value, etc.) as highlighted by informants. Therefore, understanding the sensemaking process and how these actors interpret the electrification of the forestry transportation system becomes crucial.

Interpretation of their own roles in the electrification

Actors are interpreting the new value electrified transport brings and who is willing to pay for it. The fundamental question is whether electrified transport just replaces the old transport system, or how does the value change? In turn, this can be broken down into sub-questions, e.g., how is sustainability considered? What are the working environment aspects for the drivers? How should companies handle the uncertainty of what value they offer? These issues are especially considered by the Forest companies and transport companies, particularly as the Forest Companies need to consider if customers are willing to pay for a new value. Seemingly, consumer demand for fossil-free transportation is considered low or non-existent to date, which could be seen as that they are not following a market-led approach but rather a more holistic consideration of long-term system-wide benefits (Haffar and Searcy, 2019).

Based on a long history of low margins, the transport sector is strongly focused on cost reduction, and this transformation may not fit into the current way of purchasing transportation, nor fit into how the value of products for end-consumers are marketed and valued. Interestingly, and in contrast to what Geels (2012) suggests, none of the informants in incumbent firms see their current business

model as threatened following electrification, despite new entrants and technologies. Instead, the electrified transport is a new landscape where all actors position themselves. In certain aspects, they are clear competitors, whereas in others, they identify a need to work together based on current operational needs, strategic interests, and business potential.

Providing charging is a typical question where multiple actors interpret their roles and where they could be relevant for their value proposition, creation, delivery, and capture. Seemingly, there is great uncertainty at the beginning of the transition, with several strategic and operational questions being raised. As highlighted by Alp et al. (2022), adapting electric trucks comes with the challenge of simultaneously optimizing and developing the charging infrastructure. For new actors, stand-alone charging point actors, the issues revolve around how to find a position in the value chain and build new relationships. For transporters and forest companies, this issue is ongoing, leading them to consider what role they have. Should they own and operate charging stations? What are the benefits and drawbacks? Especially when the technology is developing fast and the scale of business for electric trucks is uncertain, several actors are now finding their role in this infrastructure.

Many informants also identify a need for new actors or new competencies relating to software solutions for transport planning and charging planning. This spot and role can also be filled by different established or new actors entering the system, creating a new competitive landscape. Perhaps this uncertain development adds to the willingness of transport purchasers (i.e. the forest companies) to "take control of charging costs" by experimenting with setting up their own charging. The ones controlling the data, the planning, and charging stations are seen as a powerful new role. Depending on how they design their systems and access to charging stations, they can create lock-in situations for the ones using the charging stations. As also highlighted by Lind and Melander (2021), the actions and decisions of the truck manufacturer and how they relate to newcomer firms may be particularly interesting to study here. As also identified by informants, there are potential 'lock-in' mechanisms in play, as well as power dynamics between the actors influencing how the stakeholders in the ecosystem interpret what roles these actors play and what solutions are desirable (Geels, 2017), i.e. affecting how the dominant framing of the future transport system emerges (Kaplan, 2008).

The need to share and use data clearly increases in the learning phase for many actors to learn and operate the trucks efficiently. This need is especially apparent for the truck company, transport companies, software companies, and charging operators. Seemingly, this causes tension between project partners and throughout the value chain, and a need to address data management and data handling among actors. Who owns what data, and how can it be shared and used by others? This is clearly a crucial issue since there are differing interests among the actors in the value chain. Our informants share concerns about the need to balance openness with project partners (particularly relating to data) with protecting their own strategic interests. Here, the case of Hedvall et al. (2019) suggests that through interaction, the actors can co-create value in solutions, reduce cost, and make different combinations of solutions more efficient and valuable both to an individual firm and to other firms in the network.

Furthermore, the challenges and questions the truck manufacturer highlights concerning how they create, deliver, and capture value indicate that the electrified trucks are managed as sustainable innovation rather than an incremental change comparable to diesel (Bocken et al., 2014). Presently, they want to get new, uncertain and new technology out on the market that has a different total cost of ownership logic than traditional diesel trucks. The investment cost is higher, the fluctuation of electric price is also high but considered to be lower running costs, and there is uncertain residual value for batteries and trucks. Therefore, they need to figure out a balance to be profitable and

determine how they can support and enable trucks to enter the market on a large scale. The truck manufacturer not only has the role of enabling the trucks to enter the market but also considers what role they have in providing solutions that enable charging or swapping of batteries, as well as making transportation efficient and profitable for their customers. They also can provide data and have a role in building knowledge for learning how the new trucks operate in different conditions.

CONCLUSIONS

The use of battery-powered electric vehicles in the Swedish forestry transport industry is virtually non-existent prior to the present project. This study captures how these actors interpret the electrification in the very early stages of the transition and sheds light on implications for business model development, where the actors align, and potential areas of conflict or competition. Our findings reveal that electrification is a form of radical innovation that introduces uncertainty, disrupts long-standing practices, and requires significant learning, collaboration, and redefinition of roles among ecosystem participants. Given the early stages, the business model development is done under high uncertainty, mostly relating to short-term economic feasibility and lack of customer demand, as well as regulatory issues and market-timing. Thus, the current driving force appears not to be economically motivated but instead, the shared ambition for sustainable forestry transport.

The studied actors are engaging in sensemaking to determine their strategic positions, responsibilities in driving the shift, and business opportunities. As new roles emerge, such as those related to charging infrastructure, data ownership, and software solutions, power dynamics and potential lock-in mechanisms become central to the ecosystem's development. Ultimately, the transition to electrified forestry transport highlights the need for system-wide coordination and an emphasis on shared value creation. This study highlights the need for ecosystem-level thinking, where value creation and capture are not solely firm-centric but shaped through collaboration, shared infrastructure, and mutual learning. In this form of landscape marked by uncertainty, managers are encouraged to engage in active sensemaking around their roles, partnerships, and strategic positioning while considering power dynamics. Furthermore, the study suggests that there is a need for business model development that accommodates risk-sharing and long-term shared and sustainable value. Moreover, as the boundaries between firm's value proposition and value creation blur, especially around data ownership, charging infrastructure, and service integration, managers are faced with a challenge of balancing collaboration and retaining competitive advantage.

This study has several limitations. Firstly, the number of informants from each organization is very few, which limits the generalizability and creates risks for biases as individual informants' opinions may not be representative. Second, the data collection is in the early stages of the project, which means that the questions raised at this point in time may not be the most fundamental as the project evolves. However, this was also well in line with the ambition to capture the *initial interpretation* of electrification, rather than the entire sensemaking process. Nevertheless, future research could further examine how these dynamics unfold over time, as well as how to facilitate constructive collaborative sensemaking between the different organizations.

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